Humanitarian Logistics Hub Network Design Under Uncertainty: A Multi-Objective Model and Priority-Based Genetic Algorithm for Central Coastal Vietnam

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Abstract. Conventional disaster relief models often assume uninterrupted road networks, making them inadequate for Central Vietnam, where frequent floods severely disrupt infrastructure. We formulate the Multi-Objective Incomplete Hub Location Network Design Problem (MO-IHLNDP), a two-stage stochastic program that jointly minimizes expected logistics cost and expected maximum deprivation cost over a set of flood scenarios. The model explicitly accounts for broken inter-hub links, multi-modal transport (land, water, and air), pre-positioned inventory, as well as adaptive lateral transshipments and reactive hub activations to recover from severe localized deficits. Deprivation cost is modeled as an exponential penalty on victim waiting time, and Daganzo's continuous approximation is embedded for tractable last-mile routing cost estimation. To solve MO-IHLNDP, we propose PB-NSGA (Priority-Based Nondominated Sorting Genetic Algorithm), which confines the chromosome to first-stage hub decisions and reconstructs all scenario-dependent second-stage variables through a structured priority-based decoder, reducing search-space complexity from $O(|\mathcal{S}| \times |\mathcal{V}|)$ to $O(|\mathcal{H}|)$. On the CV-Small benchmark, PB-NSGA defines the combined reference Pareto front (normalized HV = 1.000, IGD$^+$ = 0.000) in 1.9 s – surpassing the exact Branch-and-Bound (HV = 0.895) and rendering the MILP solver infeasible within a 1800 s budget. A case study across four flood-prone provinces in Central Vietnam yields a decision-support framework: on the full-scale CV-Large instance, the model identifies the Quang Ngai Port Hub, Phuoc Son Helipad, and Tam Ky Logistics Hub as the robust investment core (100%, 70.6%, and 38.3% Pareto-selection frequency respectively), and demonstrates cross-scenario generalizability via SAA and an extreme out-of-sample "Double Typhoon" validation – both achieving zero constraint violations.

Keywords: Humanitarian logistics · Hub location · Multi-objective optimization · Stochastic programming · Disaster relief · Evolutionary algorithm

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1 Introduction

1.1 Problem

Natural disasters are becoming increasingly frequent and violent, continuously exposing supply chain weaknesses and causing massive economic losses [1]. As a highly vulnerable coastal country, Vietnam has witnessed unprecedented extreme weather events. In 2025 alone, a relentless sequence of typhoons and record-breaking floods in Central Vietnam resulted in massive fatalities and an estimated 98.7 trillion VND in economic losses [2].

Humanitarian operations during this period faced significant challenges. Most critically, national logistics infrastructure suffered severe disruption: the North-South Railway was paralyzed by track erosion and submersion, forcing the cancellation of 170 trains, while key arteries such as Vietnam National Route 1 were severed by landslides, isolating communities for days [2]. These events demonstrate that during a major catastrophe, the assumption of a fully connected transport network is fundamentally flawed – the physical network becomes “incomplete”, characterized by broken links and isolated nodes. Compounding this, resource scarcity arising from poor planning and uncertain supply and demand further strains relief efforts. Disaster Relief Network Design (DRND) bridges the preparedness and response phases by optimizing facility locations and relief routing [3].

1.2 Related Works

Humanitarian logistics and facility location. Humanitarian logistics differs fundamentally from its commercial counterpart in that speed and equity dominate over cost minimisation [4]. A comprehensive review of humanitarian facility location under uncertainty [5] identifies stochastic programming and robust optimisation as the dominant paradigms, while Boonmee et al. [6] document the growing need for models that jointly capture disruption, uncertainty, and equity. To account for social welfare, Holguín-Veras et al. [7] introduced *deprivation cost* – an exponential penalty on victim waiting time.

Hub location and incomplete networks. Hub-and-spoke structures consolidate flows efficiently and have been studied extensively since Campbell’s seminal formulations [8]. For disaster settings, the *incomplete* hub network – in which infrastructure damage severs inter-hub arcs – is a more realistic topology than the classical fully-connected graph [9]. Sangsawang and Chanta [10] proposed a pioneering flood-specific hub location model, a single-objective deterministic formulation for Thailand, establishing that hub-and-spoke structures yield measurable gains in relief distribution efficiency. Multi-modal hub networks further address the heterogeneous transport modes (road, waterway, air) that characterise disaster response [11].

Stochastic and robust optimisation for relief networks. The two-stage stochastic programming framework naturally separates pre-disaster strategic decisions from post-disaster recourse [12]. Under demand and supply uncertainty, robust formulations [13] and prepositioning models have been shown to outperform deterministic baselines. Disruption of transportation infrastructure compounds this uncertainty by rendering pre-planned routes infeasible [14], a condition endemic to Central Vietnam's flood disasters.

Multi-objective evolutionary algorithms. NSGA-II [15] is the standard framework for multi-objective combinatorial problems in humanitarian logistics, owing to its parameter-free Pareto ranking and elitist selection. Zhou et al. [16] applied a multi-objective evolutionary algorithm to multi-period dynamic emergency resource scheduling under uncertain road networks, demonstrating the effectiveness of population-based search when scenario-dependent decisions are coupled to shared strategic variables. Li et al. [11] designed a bi-objective multimodal hub-and-spoke network for emergency relief using a meta-heuristic approach, showing that problem-specific customized operators are essential for maintaining feasibility across complex network topologies.

Research gap and contributions. To the best of our knowledge, existing literature rarely simultaneously addresses incomplete hub network design, two-stage stochastic programming, multi-objective humanitarian objectives (logistics cost and deprivation cost), and a flood-disaster case study. Sangsawang and Chanta [10] is the closest precedent, yet it is deterministic and single-objective. This paper closes the gap by formulating the MO-IHLNDP and solving it via our proposed PB-NSGA, a priority-based [17] evolutionary algorithm, as this encoding strategy is proven to be particularly effective for handling the complex decision variables of two-stage stochastic programming.

2 Problem Description and Mathematical Model

We model the network design problem as a capacitated, single-allocation MO-IHLNDP. The network comprises **origins** (supply sources), **demands** (victim groups), and **hubs** of two types: planned (pre-disaster) and reactive (post-disaster, dynamically opened).

From a practical standpoint, DRND involves two critical uncertainty factors. First, infrastructure disruption – damaged roads and buildings – can disable pre-programmed plans, urging the need for a dynamic solution [13]. Second, the exact locations of demands, the total number of victims, and the availability of origins are all highly uncertain during rescue operations [12]. To account for these, we employ a two-stage stochastic programming approach across multiple scenarios, each assigning a specific risk factor per area, and assume that Search and Rescue operations rely heavily on dynamic external supply sources, as pre-positioned hubs may not fully satisfy uncertain demands under extreme scenarios.

The model integrates the deprivation cost [7] for social equity, and Daganzo's continuous approximation (CA) [18] to tractably estimate last-mile rescue times.

2.1 Sets and Indices

$\mathcal{H}, \mathcal{I}, \mathcal{J}$	Set of candidate hubs (k, h), demand nodes (i), and origins (j)
$\mathcal{S}, \mathcal{M}$	Set of disaster scenarios (s) and transportation modes (m)
$\mathcal{V}$	Set of all nodes (u, v) representing the area in question, $\mathcal{V} = \mathcal{H} \cup \mathcal{I} \cup \mathcal{J}$

2.2 Parameters

First-Phase Parameters (Strategic Planning)

F_k	Fixed cost to establish hub k of the first type (before the disaster)
c_k, κ_k	Unit holding cost and capacity limit for inventory at hub k
α, χ, M	Discount factor for economies of scale, max acceptable risk threshold, and Big-M constant
C_{uvm}, τ_{uvm}	Unit transportation cost and estimated travel time from node u to v via mode m
C_m, Q_m	Unit local routing cost and transportation capacity for vehicle mode m
γ	Conversion factor (average amount of relief items required per evacuated person)

Second-Phase Parameters (Disaster Response, Scenario s)

π_s	Probability of scenario s
F^a_{ks}	Fixed cost to establish hub k of the second type (reactive) in scenario s
τ_{ks}	Processing time for each rescue round trip at hub k in scenario s
a_{uvms}	Accessibility of arc (u, v) via mode m in scenario s (1 if traversable, 0 otherwise)
r_{us}, λ_{is}	Risk index of node u, and deprivation sensitivity coefficient for demand i in scenario s
D_{is}, O_{js}	Demand (number of people to evacuate) at i, and relief items provided at origin j in scenario s
c_{jks}	Pre-computed cheapest transport cost from origin j to hub k in scenario s: $c_{jks} = \min_{m \in \mathcal{M} \mid a_{jkms}=1} C_{jkm}$, $(+\infty$ if no available mode)
Θ_{kis}	Pre-computed CA routing cost to execute the evacuation from grid i to hub k in scenario s
Φ, η, A_i	Daganzo's circuity factor, average group size per distress location, and area of grid cell i

2.3 Decision Variables

$x_k \in \{0, 1\}$	1 if hub k is established as the first type (planned), 0 otherwise
$y_{ks} \in \{0, 1\}$	1 if hub k is established as the second type (reactive) in scenario s, 0 otherwise
$z_{iks} \in \{0, 1\}$	1 if demand i is assigned to be evacuated to hub k in scenario s, 0 otherwise
$z_{jks} \in \{0, 1\}$	1 if origin j supplies hub k in scenario s, 0 otherwise
q_k	Amount of relief items inventoried at hub k
f_{khms}	Lateral trans-shipment flow volume from hub k to hub h via mode m in scenario s

2.4 Objectives

Objective 1: Minimize Total Expected Logistics Cost (Z_1)

$$\begin{aligned}
\text{Min } Z_1 = & \underbrace{\sum_{k\in\mathcal{H}} F_k x_k + \sum_{k\in\mathcal{H}} c_k q_k}_{\text{Pre-disaster strategic costs}} \\
& + \sum_{s\in\mathcal{S}} \pi_s \Bigg[\underbrace{\sum_{k\in\mathcal{H}} F^a_{ks} y_{ks}}_{\text{Reactive setup}} + \underbrace{\sum_{j\in\mathcal{J}} \sum_{k\in\mathcal{H}} c_{jks} \cdot O_{js} z_{jks}}_{\text{Origin to Hub supply flow}} \\
& + \underbrace{\sum_{k\in\mathcal{H}} \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} \alpha C_{khm} f_{khms}}_{\text{Inter-hub lateral trans-shipment}} + \underbrace{\sum_{k\in\mathcal{H}} \sum_{i\in\mathcal{I}} \Theta_{kis} z_{iks}}_{\text{Grid-based Last-Mile CA}} \Bigg]
\end{aligned} \tag{1}$$

Objective Z_1 minimizes total expected logistics cost. Using the previously mentioned CA [18], demands are aggregated on a grid basis to pre-compute the last-mile routing cost Θ_{kis}. The first term is line-haul cost from hub k to demand i, and the second term is local detour cost inside area i:

$$\Theta_{kis} = \min_{m\in\mathcal{M}|a_{ikms}=1} \left(2 \cdot C_{kim} \cdot \lceil D_{is}/Q_m \rceil + C_m \cdot \Phi \sqrt{\lceil D_{is}/\eta \rceil \cdot A_i} \right)$$

Objective 2: Minimize Expected Maximum Deprivation Cost (Z_2)

$$\text{Min } Z_2 = \sum_{s\in\mathcal{S}} \pi_s \left(\max_{i\in\mathcal{I}} \left[D_{is} \cdot \left(e^{\lambda_{is}\cdot\Omega_{is}} - 1 \right) \right] \right) \tag{2}$$

$$\Omega_{is} = \sum_{k\in\mathcal{H}} z_{iks} \left(\tau_{ks} + 2 \cdot \min_{m\in\mathcal{M}|a_{ikms}=1} \tau_{kim} \right)$$

Objective Z_2 minimizes the expected value of maximum deprivation cost [7]. The exponential function heavily penalizes the model if it lets a demand node wait slightly longer than the rest, forcing the model to distribute resources evenly. The penalty function is adjusted by $\lambda_{is} = \lambda_0(1 + r_{is})$ so that highly vulnerable populations are prioritized.

To linearize Z_2, we exploit the fact that the model strictly enforces a single-allocation property (9). The deprivation cost for any specific $i \to k$ configuration is deterministic and finite. Thus, we can pre-compute the exact deprivation cost C^{dep}_{iks} offline:

$$C^{\text{dep}}_{iks} = D_{is} \cdot \left(e^{\lambda_{is}\cdot\left(\tau_{ks} + 2\cdot\min_{m\in\mathcal{M}|a_{ikms}=1} \tau_{kim}\right)} - 1 \right) \tag{3}$$

By introducing a continuous auxiliary variable $W_s \geq 0$ to capture the "worst-case" (maximum) expected deprivation cost among all demand nodes in a given scenario s, the non-linear Z_2 objective can be reformulated into an exact linear sum:

$$Z_2^{\text{linear}} = \sum_{s\in\mathcal{S}} \pi_s \cdot W_s \tag{4}$$

Subject to:

$$W_s \geq \sum_{k\in\mathcal{H}} C_{iks}^{\text{dep}} \cdot z_{iks} \quad \forall i \in \mathcal{I}, s \in \mathcal{S} \tag{5}$$

2.5 Constraints

The complete MO-IHLNDP model is formally stated as follows:

$$\begin{aligned}
&\text{Minimize} \quad Z_1, Z_2 \\
&\text{subject to} \quad \text{Constraints } (6) - (19)
\end{aligned}$$

$$x_k + y_{ks} \leq 1 \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{6}$$

$$q_k \leq \kappa_k \cdot x_k \quad \forall k \in \mathcal{H} \tag{7}$$

$$r_{ks} \cdot (x_k + y_{ks}) \leq \chi \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{8}$$

$$\sum_{k\in\mathcal{H}} z_{iks} = 1 \quad \forall i \in \mathcal{I}, s \in \mathcal{S} \tag{9}$$

$$\sum_{k\in\mathcal{H}} z_{jks} = 1 \quad \forall j \in \mathcal{J}, s \in \mathcal{S} \tag{10}$$

$$z_{iks} \leq x_k + y_{ks} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \tag{11}$$

$$z_{jks} \leq x_k + y_{ks} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \tag{12}$$

$$z_{iks} \leq \sum_{m\in\mathcal{M}} a_{ikms} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \tag{13}$$

$$z_{jks} \leq \sum_{m\in\mathcal{M}} a_{jkms} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \tag{14}$$

$$f_{khms} \leq M \cdot a_{khms} \quad \forall k, h \in \mathcal{H},\ m \in \mathcal{M},\ s \in \mathcal{S} \tag{15}$$

$$\begin{aligned}
\sum_{i\in\mathcal{I}} \gamma D_{is} z_{iks} + \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} f_{khms} \leq q_k + \sum_{j\in\mathcal{J}} O_{js} z_{jks} \\
+ \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} f_{hkms} \quad \forall k \in \mathcal{H}, s \in \mathcal{S}
\end{aligned} \tag{16}$$

$$q_k + \sum_{j\in\mathcal{J}} O_{js} z_{jks} + \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} f_{hkms} \leq \kappa_k \cdot (x_k + y_{ks}) \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{17}$$

$$x_k, y_{ks}, z_{iks}, z_{jks} \in \{0, 1\} \quad \forall i, j, k, s \tag{18}$$

$$q_k, f_{khms}, W_s \geq 0 \quad \forall k, h, m, s \tag{19}$$

Constraints (6) define the hub establishment. For planned hubs, inventory is bounded by physical capacity (7), while reactive hubs are required to be placed within safe zones (8). Constraints (9)-(14) enforce the single-allocation property to active hubs via available links. Constraint (15) restricts flow to be trans-shipped on intact paths. Constraints (16) and (17) maintain flow conservation and capacity limits. Specifically, (16) ensures that total supply, comprising inventory (q_k), external origins (O_{js}), and incoming flows – satisfies both victim demand (converted by γ) and outgoing trans-shipments. Variable domains are defined in (18)-(19).

Constraints (5) define the linear boundaries for the maximum deprivation cost auxiliary variables W_s. The linearized program is then:

$$\begin{aligned} &\text{Minimize} && Z_1, Z_2^{\text{linear}} \\ &\text{subject to} && \text{Constraints } (5)-(19) \end{aligned} \tag{20}$$

3 Proposed Algorithm: PB-NSGA

The MO-IHLNDP is NP-hard, as its single-objective, single-scenario relaxation subsumes the uncapacitated facility location problem [19]. We therefore propose **PB-NSGA** (Priority-Based Nondominated Sorting Genetic Algorithm), which embeds the NSGA-II framework [15] with a custom heuristic decoder.

3.1 Chromosome Encoding Scheme

Each chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{A}, \mathbf{W})$ consists of four segments. The binary vector $\mathbf{X} \in \{0,1\}^{|\mathcal{H}|}$ determines planned hub establishment. The continuous vector $\mathbf{R} \in [0,1]^{|\mathcal{H}|}$ defines the inventory fill fraction, setting $q_k = R_k \cdot \kappa_k$. The integer vector $\mathbf{A} \in \{0,\ldots,|\mathcal{H}|-1\}^{|\mathcal{I}|}$ encodes the preferred anchor hub for each demand i. The continuous vector $\mathbf{W} \in [0,1]^6$ contains six heuristic weights governing the decoder: w_0 and w_3 dictate demand sorting priority by urgency ($\lambda \cdot D$) and isolation; w_1, w_2, w_4 score candidate hubs by travel time, residual inventory, and preference for planned over reactive hubs; w_5 controls conservatism towards opening new reactive hubs.

3.2 Priority-Based Decoding Heuristic

The decoder is invoked once per chromosome per scenario $s \in \mathcal{S}$, reconstructing all second-stage decisions from $\mathbf{C}$ in four steps.

Step 1 – Initialisation. Planned hubs with $X_k = 1$ and $r_{ks} \leq \chi$ are activated with pre-positioned inventory $q_k = R_k \cdot \kappa_k$. If none qualify, the safest hub is forced active.

Step 2 – Demand Prioritisation. Each demand node i receives a composite score prioritising urgency, isolation, and proximity to active hubs. Demands are then processed in descending score order.

Step 3 – Tiered Hub Assignment. For each demand i, active hubs are trialled in proximity order from the anchor hub π_{A_i}. *Pass 1* selects the best candidate by residual capacity, travel time, and operational status. If no feasible candidate is found, *Pass 2* relaxes capacity constraints. If no reachable active hub exists, the decoder opens the nearest safe inactive hub as a reactive fallback and increments CV.

Step 4 – Supply Routing and Objective Accumulation. Origins supply the most under-supplied active hubs, and surplus is redistributed via discounted inter-hub trans-shipment (α). Expected objectives Z_1 and Z_2 are accumulated across all $s \in \mathcal{S}$. Algorithm 1 formalises these procedures.

Algorithm 1 Priority-Based Decoder (single scenario s)

Require: Chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{A}, \mathbf{W})$, scenario s, instance data
Ensure: $(Z_1^s,\ Z_2^s,\ \mathrm{CV}^s)$
Activate $\mathcal{H}_{\mathrm{act}} \leftarrow \{k : X_k = 1,\ r_{ks} \le \chi\}$; set $q_k \leftarrow R_k \kappa_k$
if $\mathcal{H}_{\mathrm{act}} = \emptyset$ **then**
    Force-activate $\arg\min_k r_{ks}$ among $\{k : X_k = 1\}$
end if
Score demands by urgency, isolation, and proximity; sort descending
for each demand i (in priority order) **do**
    $K \leftarrow \max(1, \lceil w_5 |\mathcal{H}| \rceil)$; trial order $\leftarrow \pi_{A_i}$
    Pass 1: $k^* \leftarrow \arg\max \mathrm{HubScore}(k)$ over first K active, reachable hubs with residual > 0
    if Pass 1 succeeds **then**
        Assign $i \to k^*$; deduct demand from residual inventory of k^*
    else if any active reachable hub remains **then**
        Pass 2: assign i to first such hub (capacity violation allowed)
    else
        Reactive fallback: open nearest safe inactive hub; add F^a_{ks}; $\mathrm{CV}^s \mathrel{+}= 1$
    end if
end for
Assign origins greedily; redistribute surplus via inter-hub trans-shipment.
Accumulate Z_1^s, Z_2^s
return $(Z_1^s,\ Z_2^s,\ \mathrm{CV}^s)$

3.3 Genetic Operators and Selection

Selection uses binary tournament on Pareto rank with crowding distance as tiebreaker. To prevent stagnation in the multi-modal search space, our implementation monitors the diversity of the rank-1 front in objective space (Z_1, Z_2). If the front remains unchanged for 20 generations, selection pressure is boosted via 3-way tournaments and weight-vector (W) hypermutation is applied to escape the local basin. *Crossover* applies Uniform Crossover to $\mathbf{X}$ and $\mathbf{A}$, and SBX [15] to the continuous segments $\mathbf{R}$ and $\mathbf{W}$, whose distribution-index control concentrates offspring near parents and preserves the parental mean on bounded real vectors. *Mutation* ($p_m = 1/|\mathbf{C}|$) applies Bit-flip to $\mathbf{X}$, random-reset to $\mathbf{A}$, and Polynomial mutation to $\mathbf{R}$ and $\mathbf{W}$. Constraint handling follows the constrained-dominance principle [15]: feasible solutions always dominate infeasible ones; among infeasible solutions, lower CV dominates. When a partial Pareto front must be split, tiebreaking proceeds by (1) Pareto rank, (2) crowding distance, (3) minimum Hamming distance in $\mathbf{X}$-space – preserving genotypic diversity in hub configuration and countering premature convergence.

4 Computational Experiments

4.1 Experimental Setup

PB-NSGA is implemented in C++17; dataset construction and post-processing are performed in Python 3, on a workstation with an 11th Gen Intel Core i5-1135G7 @ 2.40 GHz and 16 GB RAM. Algorithm parameters are fixed for all runs: $N = 200$, $G = 300$, $p_c = 0.9$, $\eta_c = \eta_m = 20$, $p_m = 1/|\mathbf{C}|$. Problem constants are $\chi = 0.7$, $\alpha = 0.6$, $\gamma = 3.0$ kg/person, $\lambda_0 = 0.8$. Each configuration is executed for 20 independent runs, reporting mean ± std. We use **Hypervolume (HV)** [20] and **IGD$^+$** [21] for performance metrics. Pareto union across all algorithms served as a proxy ground truth for small instances.

4.2 Dataset Description

Central Vietnam Dataset. We construct two instances modelling four flood-prone provinces: Da Nang, Quang Nam, Thua Thien-Hue, and Quang Ngai. **CV-Small** uses district-level demand nodes ($|\mathcal{I}| = 20$, $|\mathcal{H}| = 5$, $|\mathcal{J}| = 2$, $|\mathcal{S}| = 3$); **CV-Large** uses commune-level resolution ($|\mathcal{I}| = 100$, $|\mathcal{H}| = 20$, $|\mathcal{J}| = 12$, $|\mathcal{S}| = 3$). Node coordinates map to real administrative centroids, confirmed logistics facilities, and vital supply origins. Scenario risks and road disruptions are calibrated using a multi-criteria flood vulnerability score [22]. Three transport modes – trucks, motorboats, and helicopters – are considered. Roads are subject to disruption, while water mode is enabled on flood areas. Helicopter links (air mode) are capped at 15% of all active routing connections per scenario, approximating a shared rotary-wing fleet of 2–3 aircraft across the 20 candidate hubs.

4.3 Experiment 1: Baseline Comparison

To evaluate PB-NSGA against the structural complexity of MO-IHLNDP, we benchmarked the algorithm against two baselines on the risk-calibrated **CV-Small** sub-instance ($|\mathcal{H}| = 5, |\mathcal{I}| = 20$). The first baseline is a two-phased **Greedy Heuristic** that constructs upper bounds by selecting the cheapest or safest hubs and assigns demands via nearest-neighbour logic. The second is an exact **Mixed-Integer Linear Programming (MILP)** formulation solved via OR-Tools [23] with the ϵ-constraint method.

Table 1 reveals an instructive pattern for the small instance. The greedy heuristic remains distant from the reference front despite stochastic resets. The

Table 1. Baseline comparison on the CV-Small instance. HV is normalized w.r.t. the combined non-dominated reference front (BB ∪ Greedy ∪ PB-NSGA). −: no feasible solution returned.

Algorithm	HV (norm.) ↑	IGD$^+$ ↓	CPU Time (s) ↓
Greedy Heuristic	0.000	1.538	< **0.1**
MILP (ϵ-constraint)	−	−	1800.0*
Exact Enum (BB)	0.895	0.079	0.3
PB-NSGA (Ours)	**1.000**	**0.000**	1.9

* No feasible Pareto solution found within 1800 s (SCIP/Big-M instability).

MILP ϵ-constraint solver failed to return any feasible Pareto solution within 1800 s, consistent with known Big-M numerical difficulties on non-linear deprivation objectives. PB-NSGA converges to the optimal region and defines the combined non-dominated reference front, reaching a hypervolume of 1.000. This efficiency stems from PB-NSGA's ability to simultaneously optimize the decoder's priority-weight vectors (W), discovering more refined routing and inventory patterns for each hub configuration than the exhaustive BB enumeration with fixed weights.

As demonstrated in Experiment 2 (Sec. 4.4), this scalability advantage becomes even more pronounced on larger instances where exact methods become computationally intractable.

4.4 Experiment 2: Deep Case Study Analysis & SAA Verification

To investigate the practical managerial insights generated by the model on realistic operational scales, we executed PB-NSGA on the **CV-Large** instance ($|\mathcal{I}| = 100, |\mathcal{H}| = 20$). Table 2 reports the algorithm's performance metrics.

Table 2. PB-NSGA performance metrics across 20 independent runs (mean ± std). #Pareto (combined) is the size of the union Pareto front across all runs.

Instance	HV (norm.) ↑	IGD$^+$ ↓	#Pareto (mean)	#Pareto (combined)
CV-Small ($\lvert\mathcal{I}\rvert$=20)	1.133 ± 0.084	0.092 ± 0.075	21.1 ± 34.1	6
CV-Large ($\lvert\mathcal{I}\rvert$=100)	0.960 ± 0.090	0.117 ± 0.051	10.7 ± 3.8	17

Table 2 demonstrates that PB-NSGA maintains competitive solution quality even as the problem scales to commune-level resolution ($2^{20} \approx 10^6$ hub configurations). On the CV-Large instance, the algorithm achieves a mean IGD$^+$ of 0.117 ± 0.051 and a combined Pareto front of 17 non-dominated solutions across all 20 runs, indicating consistent convergence toward a stable region of the objective space. The HV standard deviation (0.090) reflects the inherent stochasticity of searching a vast multimodal landscape, yet remains within acceptable bounds for a practical decision-support tool.

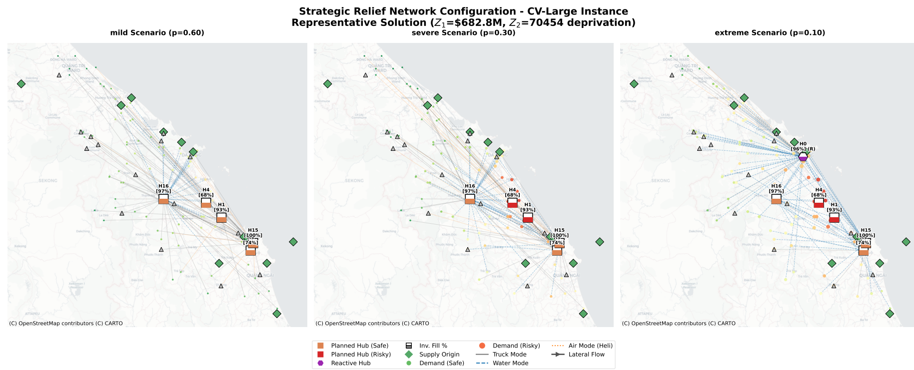

Fig. 1. Detailed multi-modal network mapping for a representative “Balanced” trade-off solution in CV-Large across three probabilistic scenarios (Mild, Severe, Extreme). Modes are indicated by distinct line styles and colors.

Figure 1 visualizes a representative “Balanced” solution. Under mild conditions, the network utilizes a dense combination of coastal watercraft links and terrestrial road networks, drawing heavily from pre-positioned inventory at strategic hubs H4, H15, and H16 (indicated by their high filling ratios). As severity intensifies to the extreme scenario, a pronounced modal shift occurs: the decoder activates rotary-wing (helicopter) links to reach isolated clusters in the mountainous north-west and vulnerable eastern regions. To supplement critical logistics bottlenecks, the model triggers a reactive hub (H8, shown in purple) and facilitates lateral transshipment flows between established hubs to rebalance inventory deficits.

Sample Average Approximation and Out-Of-Sample Validation To evaluate whether PB-NSGA’s solutions generalize beyond the three training scenarios, we validated the CV-Large Pareto front against two Out-Of-Sample tests.

First, we generated a Sample Average Approximation (SAA) set of 100 scenarios with randomized geographic flood risk decay and stochastic epicenter placements. Evaluating PB-NSGA’s first-stage hub decisions and trained weights $\mathbf{W}$ on this unseen set yielded 0.00 expected Constraint Violations (CV), confirming operational feasibility across a dense distribution of disaster realizations.

Second, we confronted the trained model with an extreme Out-Of-Sample (OOS) “Double Typhoon” event – a catastrophic scenario featuring four simultaneous epicenters and a 3.5× severity threshold outside the training distribution. PB-NSGA’s stored priority-allocation strategy absorbed this shock with 0 CV across all Pareto solutions, indicating that the learned hub-choice anchors $\mathbf{A}$ and priority weights $\mathbf{W}$ encode a robust emergency response policy rather than overfitting to training sample geometry.

Methodological Framework for Decision Support *Robust hub prioritization.* Across all 20 independent runs, hub selection frequency on the CV-Large combined Pareto front reveals a clear hierarchy of strategic priority. The Quang Ngai Port Hub appears in every Pareto-optimal solution (100% selection frequency), establishing it as the unconditional anchor of the relief network. The Phuoc Son Helipad follows at 70.6%, and the Tam Ky Logistics Hub at 38.3%, together forming a robust hub core for permanent infrastructure investment. Hubs that are cost-effective under mild conditions but breach the $\chi = 0.7$ safety threshold during extreme disruptions expose a critical *resilience gap*: identifying these gaps enables authorities to pre-position backup inventory or activate contingency evacuations before a disaster strikes.

Transport mode diversification. The extreme scenario enforces a pronounced modal shift away from disrupted road networks. Pre-equipping coastal hubs – particularly the robust core anchored by the Quang Ngai Port and Phuoc Son Helipad – with resilient watercraft terminals and maintaining emergency airspace access for rotary-wing aircraft delivers asymmetric resilience yield at a fraction of the cost of deploying redundant mega-hubs.

5 Conclusion and Future Work

This paper presented a two-stage stochastic multi-objective hub network design model for flood disaster relief – MO-IHLNDP – and a tailored PB-NSGA solver that jointly minimizes logistics cost and deprivation cost while explicitly handling infrastructure disruption, multi-modal transport, and pre-positioned inventory. Computational results demonstrate that PB-NSGA outperforms both exact and heuristic baselines on verifiable instances, and scales to commune-level resolution in practical runtimes. Applied to four flood-prone provinces in Central Vietnam, the framework reveals a quantified cost-equity trade-off and a stable hierarchy of critical facility investments – anchored by geographically secure coastal and highland logistics nodes – validated to generalize under unseen and catastrophic disaster scenarios. The adaptive lateral transshipment and reactive hub mechanisms proved key to maintaining network feasibility when pre-planned infrastructure is overwhelmed.

Limitations. The absence of a publicly available real-world relief dataset for Central Vietnam required the use of a synthetic instance calibrated from administrative and geographic data. Field validation against actual disaster records remains necessary before operational deployment.

Future work will prioritize four directions. First, constructing a field-validated relief logistics dataset for Central Vietnam in collaboration with local authorities. Second, integrating the model with government relief planning systems for rapid practical deployment. Third, benchmarking PB-NSGA against established multi-objective metaheuristics (MOEA/D, NSGA-III) on the CV-Large instance to further quantify the advantage of the problem-specific decoder. Fourth, strengthening PB-NSGA through adaptive operator tuning and hybridization with local search for larger-scale instances.

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